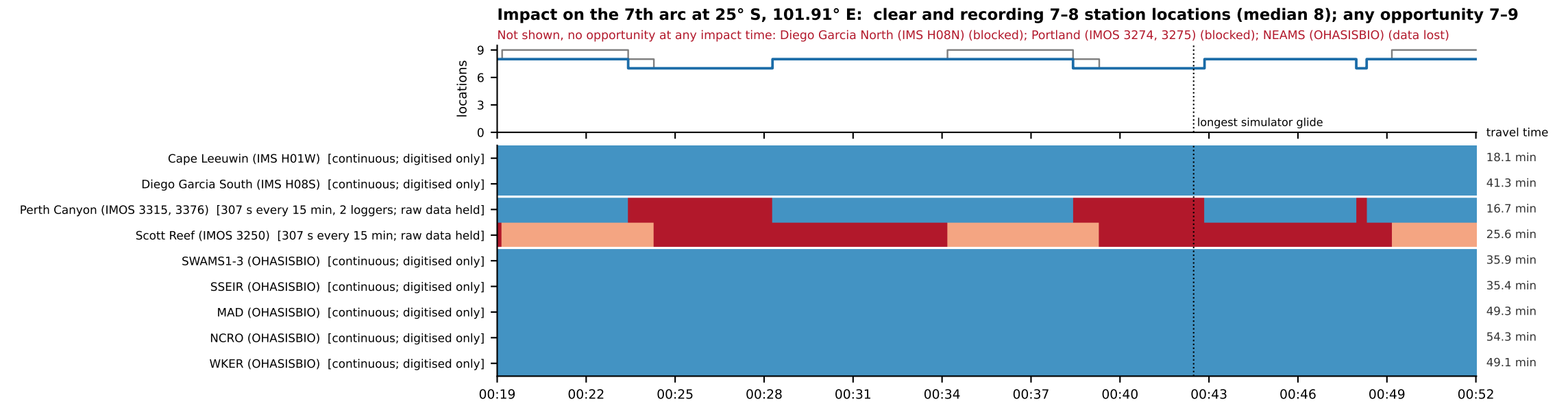
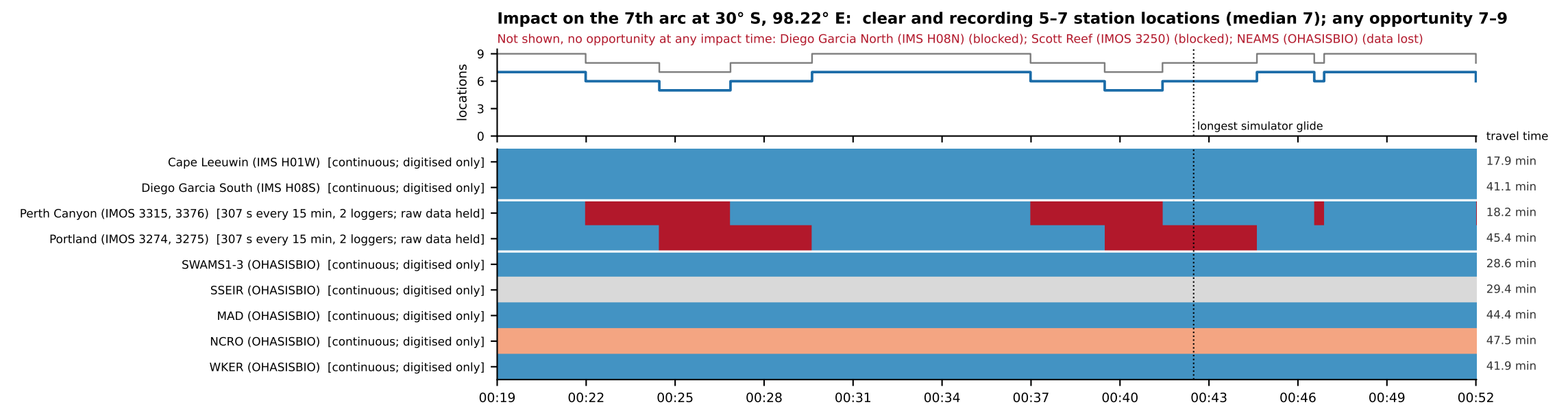
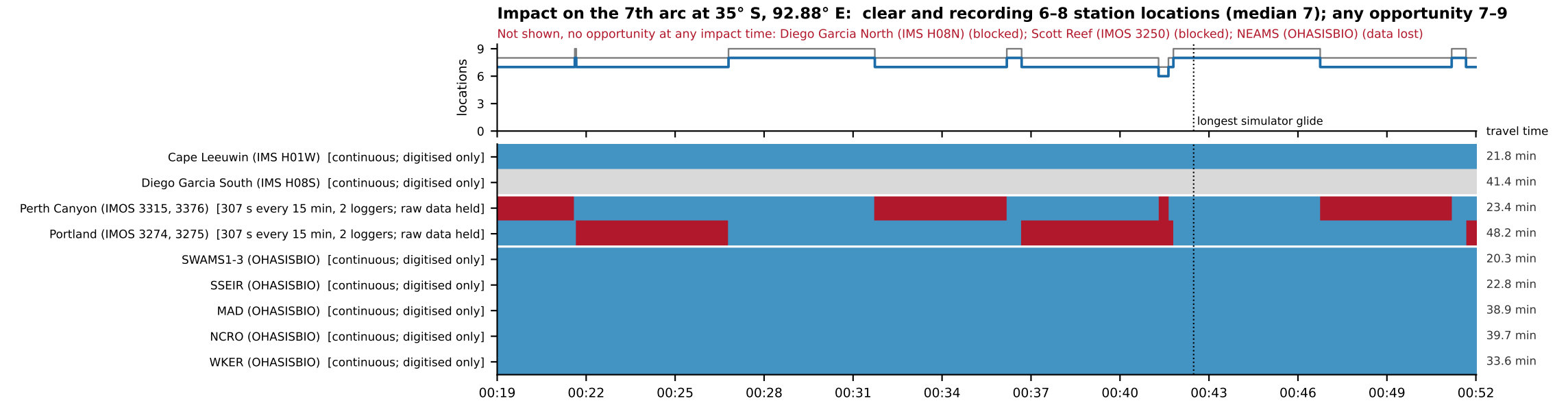
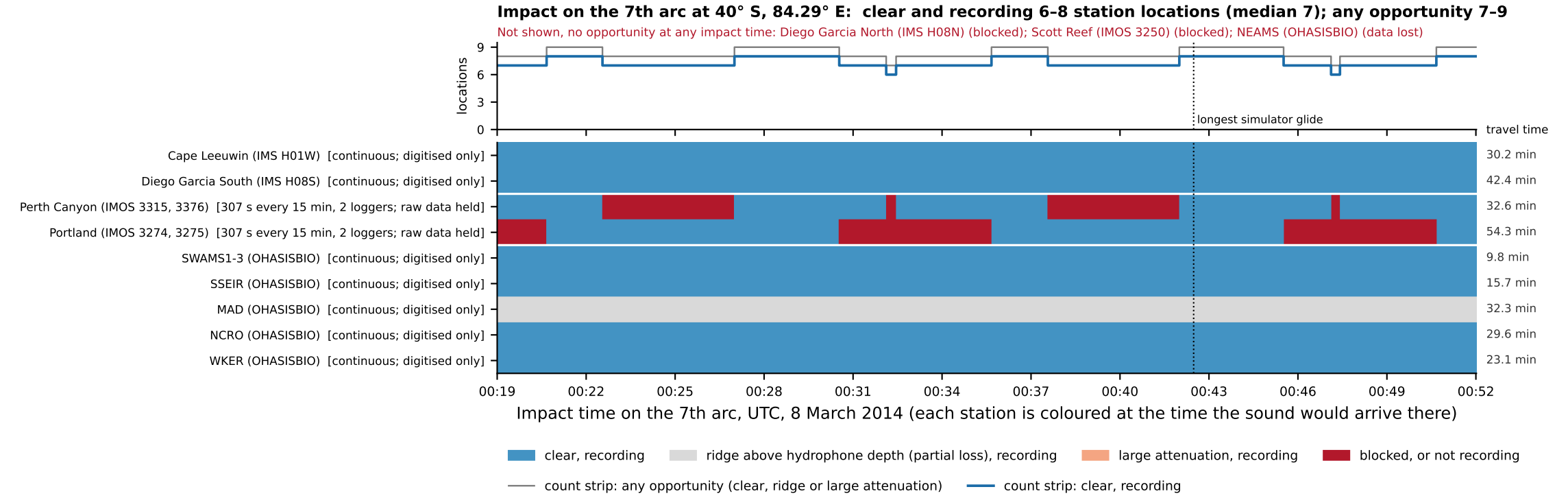
Recording opportunity by impact time: which stations had line of sight and were recording when the sound would arrive, for an impact on the 7th arc at 25° S, 30° S, 35° S and 40° S









Plain-language note. Read across from an impact time. For each station, the colour shows the state at the time the sound would arrive there (impact time plus travel time; the travel time is at the right). Red: the path is blocked, or the station was not recording at that time. Stations that have no opportunity at any impact time are not shown; they are named in red under each panel title. The Australian seabed loggers record 307 s in each 15 min, so their rows have gaps; Perth Canyon and Portland have two loggers that start 5 min apart. This shows where a sound can be received; it does not show that it was loud enough to detect. That is the next step.

Technical note. Line of sight: hydro_data.sqlite table arc_path (arc_line_of_sight.py; WGS84 geodesics on GEBCO 2026, 15 arc-s). Blocked: land or water shallower than min(receiver depth, 200 m) more than 20 km from the receiver. Large attenuation: >= 10 km shallower than 1,000 m more than 100 km from the receiver. Ridge: >= 10 km shallower than the hydrophone (deep receivers). These thresholds are agent choices, not ruled. Recording: IMOS record windows from table segment (UTC, clock-corrected, event-day and background sets joined; no record missing 00:00-05:00 UTC); OHASISBIO continuous at 240 Hz (Royer 2016), NEAMS lost during recovery; IMS H01W/H08S continuous 250 Hz (8 Mar 2014 data used in Kadri 2024, CMST 2014); H08N continuous assumed (no published use of its 8 Mar 2014 data found). A location takes its most favourable instrument that is recording. Travel time = geodesic distance / c, c = 1.482 ± 0.006 km/s (registry constant c_sofar); 1-sigma timing error 4-16 s. Not included: HA04 Crozet (installed Dec 2016), IMOS deployments with no 8 Mar 2014 files, island seismometers. Geometric screen, not transmission loss; noise (for example the seismic survey shots Royer reports) is not included. Impact time axis 00:19:00-00:52:00 UTC at 1 s steps (agent choice: covers the run J 99.9 % impact-time quantile after 00:19 at all four latitudes for the 00:19 range-and-frequency option); dotted line 00:42:29 = longest Boeing-simulator glide (23 min after 00:19:29). Run J impacts after 00:19 within 0.5° of each panel latitude, 99.9 % weighted quantile: 25° S 00:49:26 (00:19 range and frequency used), 00:51:30 (00:19 data not used); 30° S 00:48:51 (00:19 range and frequency used), 00:50:56 (00:19 data not used); 35° S 00:49:24 (00:19 range and frequency used), 00:50:27 (00:19 data not used); 40° S 00:51:34 (00:19 range and frequency used), 01:13:20 (00:19 data not used). Later impacts are not shown. Arrival treated as an instant; a sound that lasts some tens of seconds can still be partly recorded when it starts just before a record window.